Lecture 16

Large Sample Theory

Part 1 – LLN

① Let $X_1, \ldots, X_n$ be independent observations on X. Assume X has mean μ and variance σ^2.

- Recall that $\bar{X}_n = \frac{1}{n}(X_1 + \ldots + X_n)$ denotes the sample mean.

- Recall also that:

② $$E(\bar{X}_n) = \mu$$
$$Var(\bar{X}_n) = \frac{\sigma^2}{n}$$

- Important: ② holds always in the context of ①. In particular, we do not need to specify the p.d.f/p.f. of X.

General question: How close is $\bar{X}_n$ to μ?

- To offer an answer we would need to evaluate

  ③ $\mathbb{P}(|\bar{X}_n - \mu| \geq t)$, for some $t > 0$ of interest.

- Showing that $\bar{X}_n$ is close to μ reduces to showing that the probability in ③ is small.

- If we know the distribution of $\bar{X}_n$, we can calculate ③ exactly (see Example 6.1.1.b).

- In general, the distribution of $\bar{X}_n$ is (may be) difficult to calculate.

General strategy for showing that ③ is small

(a) Find a bound on ③.

(b) Show that the bound is small.

- To address (a) we have many possibilities. For this course we will only make use of

Chebyshev's inequality For any r.v. X s.t. $\text{var}(X) > 0$, and for any $t > 0$, we have:

(4) $\mathbb{P}(|X - E(X)| \geq t) \leq \frac{\text{var}(X)}{t^2}$.

(See Th. 6.2.2. for a proof.)

- Quantifying how close $\bar{X}_n$ is to μ, in probability

Combining (2), (3) and (4) we have:

(5) $\mathbb{P}(|\bar{X}_n - \mu| \geq t) \leq \frac{\sigma^2}{n \cdot t^2}$.

Remark 1: As n gets larger (that is, as we accumulate more observations), the RHS in (5) gets smaller.

- Examples 6.2.1, 6.2.3 text book

Example 6.2.1 Determining the Required Number of Observations. Suppose that a random sample is to be taken from a distribution for which the value of the mean μ is not known, but for which it is known that the standard deviation σ is 2 units or less. We shall determine how large the sample size must be in order to make the probability at least 0.99 that $|\overline{X}_n - \mu|$ will be less than 1 unit.

Solution: We need to find n s.t. $\mathbb{P}(|\overline{X}_n - \mu| \leq 1) \geq 0.99$. Equivalently, we need n s.t: $1 - \mathbb{P}(|\overline{X}_n - \mu| \leq 1) \leq 1 - 0.99 \iff \mathbb{P}(|\overline{X}_n - \mu| \geq 1) \leq 0.01$.

Since $\sigma^2 \leq 2^2 = 4$, it follows from the relation (6.2.4) that for every sample size n,

$$\Pr(|\overline{X}_n - \mu| \geq 1) \leq \frac{\sigma^2}{n} \leq \frac{4}{n}.$$

Since n must be chosen so that $\Pr(|\overline{X}_n - \mu| < 1) \geq 0.99$, it follows that n must be chosen so that $4/n \leq 0.01$. Hence, it is required that $n \geq 400$.

The Law of Large Numbers

Definition: A sequence $Z_1, Z_2, \ldots, Z_n \ldots$ of random variables converges to b in probability if for every $t > 0$ we have

$$\lim_{n \to \infty} \mathbb{P}(|Z_n - b| < t) = 1.$$

- We write $Z_n \xrightarrow[n \to \infty]{P} b$; "$Z_n$ converges in probability to b, as n converges to ∞."

Theorem $\boxed{\overline{X}_n \xrightarrow[n \to \infty]{P} \mu}$, where $\overline{X}_n = \frac{X_1 + \ldots + X_n}{n}$, the X_i's are independent with mean μ and $\sigma^2 > 0$.

Proof From ⑤:

$$\mathbb{P}(|\overline{X}_n - \mu| < t) \geq 1 - \frac{\sigma^2}{nt^2}$$

$$\Downarrow$$

$$\lim_{n \to \infty} \mathbb{P}(|\overline{X}_n - \mu| < t) = 1.$$

Q.e.d.

Exercise 5, page 359, textbook

- Let $X_1, \dots, X_n$ be i.i.d. from a distribution with mean μ and standard deviation σ.
- Determine the value of n such that

$$\mathbb{P}(|\bar{X}_n - \mu| \leq 2\sigma) \geq 0.99.$$

Solution Equivalently, we will find n such that:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq 2\sigma) \leq 0.01.$$

By Chebyshev's inequality, we always have:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq 2\sigma) \leq \frac{\sigma^2}{n \cdot 2^2 \cdot \sigma^2} = \frac{1}{4n}.$$

We therefore need to find n such that

$$\frac{1}{4n} \leq \frac{1}{100} \iff 100 \leq 4n \iff \boxed{n \geq 25}$$

Very important results

① If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$,
then $X_1 + \dots + X_n \sim N(n\cdot\mu, n\cdot\sigma^2)$.

② If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$,
then $\boxed{\overline{X}_n =: \frac{X_1 + \dots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right).}$

Moral: • If "the data" (that is, $X_1, \dots, X_n$) are exactly $N(\mu, \sigma^2)$ random variables, then $\overline{X}_n$ is "centered around μ" in the sense discussed in the context of the Gaussian distribution.

• If "the data" are not necessarily $N(\mu, \sigma^2)$, the LLN tells us that if n is large (that is, we have enough data), $\overline{X}_n$ is still "close" to μ.